

Supplementary Material: Local Refinement with Frozen Estimator Applications

Internal draft for author review

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1 Status and reading guide

Tables and figures copied from the original manuscript describe historical YOLO experiments. The DOPE and ResNet18 extensions are identified separately below; no value is filled from another estimator. This supplement preserves the distinction between completed reused-development evidence and independent confirmation. All three matched-budget fits were completed; the learning curves are diagnostic, not a best-checkpoint selector. Independent real-session confirmation is not yet available.

The completed ResNet18 FULL estimator uses RGB and five registered-dimension values jointly. Historical YOLO/DOPE heads remain image-feature based and use dimensions only in PnP. ResNet P0 is the primary third-backbone transfer arm; P5/P5_CONSTANT separately tests a dimension input to the correction head. These roles must not be merged into a single dimension-benefit claim.

2 Historical YOLO development results

Original-pixel medians and P90 are pooled over each model’s supervised matched points. Frame mean is the mean of matched per-frame errors, not the pooled median. Gross20 and PCK are percentages; PCK retains the full 2,818-point ground-truth denominator. Raw prior rows, when available, precede no claim that the selected capped wrapper is the original standalone method.

Model	Med	P90	Mean	Gross20	PCK5	PCK10	PCK20	N
R0	6.616	38.670	21.673	17.20	37.19	63.73	80.98	2756
P1	5.851	38.025	20.930	15.57	42.26	67.64	82.58	2756
P2	5.943	37.728	21.010	15.89	42.16	67.07	82.26	2756
P3	5.921	37.546	20.949	15.42	41.66	67.32	82.72	2756
L1	6.052	37.204	21.145	16.07	41.23	66.57	82.08	2756
L2	6.037	37.302	21.085	15.93	41.06	66.93	82.22	2756
L3	6.121	37.453	21.164	16.00	41.02	66.86	82.15	2756
D1	6.366	37.479	21.411	16.11	39.14	64.66	82.04	2756
D2	6.457	37.810	21.456	16.44	38.75	64.62	81.72	2756
D3	6.467	37.410	21.435	16.22	38.96	64.44	81.94	2756
PRIOR1	5.569	37.218	20.784	15.49	44.78	68.31	82.65	2756
PRIOR1_raw	5.569	36.776	20.655	15.42	44.78	68.28	82.72	2756
PRIOR2	5.541	37.373	20.738	15.64	44.75	68.84	82.51	2756
PRIOR2_raw	5.541	37.295	20.594	15.53	44.78	68.88	82.61	2756
PRIOR3	5.596	37.089	20.830	15.71	43.90	68.06	82.43	2756
PRIOR3_raw	5.599	36.721	20.700	15.60	43.90	68.10	82.54	2756

3 Geometry-reference results

Rotation and yaw are degrees; translation is centimeters. These are not independent physical 6D measurements. Pose coverage is separate from box matching: the canonical prediction-only selector may return a pose from a top detection that does not meet the 2D matching gate. No new symmetry minimization was introduced to excuse near-square axis confusion.

Model	Rot(deg)	Yaw(deg)	Trans(cm)	IoU3D	ADD AUC	Coverage
R0	2.262	1.231	7.897	0.603	0.428	1.000
P1	2.089	1.074	6.865	0.642	0.454	1.000
P2	2.106	1.136	6.917	0.637	0.456	1.000
P3	2.073	1.156	7.678	0.637	0.454	1.000
L1	2.110	1.115	7.609	0.637	0.450	1.000
L2	2.103	1.122	7.364	0.636	0.452	1.000
L3	2.136	1.132	7.672	0.625	0.445	1.000
D1	2.201	1.192	7.570	0.621	0.441	1.000
D2	2.181	1.182	7.540	0.619	0.442	1.000
D3	2.188	1.192	7.682	0.616	0.440	1.000
PRIOR1	2.018	1.100	7.185	0.614	0.455	1.000
PRIOR1_raw	2.018	1.100	7.185	0.616	0.456	1.000
PRIOR2	2.004	1.102	6.675	0.628	0.467	1.000
PRIOR2_raw	2.010	1.105	6.844	0.628	0.464	1.000
PRIOR3	2.132	1.190	6.825	0.619	0.460	1.000
PRIOR3_raw	2.132	1.190	6.825	0.619	0.460	1.000

4 Registered geometry and camera inputs

Object	Long(m)	Short(m)	Height(m)	Frames
Plastic	1.30	1.10	0.11	194
Wood	0.80	0.59	0.14	125

Width	Height	f_x	f_y	c_x	c_y	Frames
640	480	605.906	605.970	317.596	256.292	28
640	480	608.298	607.846	326.401	239.547	146
640	480	614.184	614.313	329.280	234.529	100
1280	720	908.860	908.955	636.394	384.438	45

Intrinsics above are annotation records in pixels, not newly measured calibration accuracy. The canonical solver passes no distortion coefficients. The records inspected here provide resolution and intrinsics, not an independently verified lens/calibration uncertainty report. Object dimensions are the unchanged registered geometry; the evaluation contract does not claim new physical inspection. The declared long-axis equivalence allows a half turn, but not a quarter turn. The wood object is not established as visually symmetric under the half turn. These benchmark conventions do not certify interchangeable physical fork-entry geometry.

5 Complete timing summaries

The expanded panel was measured in the same session for all comparators.

All values are milliseconds. The same-session expanded panel is used after the prior evaluation is complete; otherwise only the explicitly historical timing is available. Every measured repeat is retained. Model and image loading are excluded, but actual RGB preprocessing, detector forward, feature sampling/refinement, restoration and canonical PnP are included in their named intervals. Allocated-memory figures do not include unrelated desktop processes.

Model	2D med	mean	P90	Total med	mean	P90	Add2D med
R0	9.733	10.017	11.803	11.217	11.522	13.298	0.000
P1	13.661	14.031	15.730	15.352	15.496	17.096	4.056
P2	13.622	13.996	15.886	15.280	15.465	17.332	3.802
P3	13.784	14.096	15.984	15.550	15.596	17.355	4.077
L1	18.226	18.106	20.105	19.736	19.598	21.496	7.975
L2	18.150	18.163	20.161	19.858	19.670	21.572	8.115

Model	2D med	mean	P90	Total med	mean	P90	Add2D med
L3	18.178	18.194	20.097	19.734	19.685	21.457	8.194
D1	13.231	13.509	15.482	14.800	15.022	16.941	3.343
D2	12.979	13.464	15.558	14.800	14.993	16.948	3.284
D3	13.226	13.444	15.365	14.773	14.927	16.832	3.211
PRIOR1	27.537	27.380	28.439	29.020	28.926	29.890	17.466
PRIOR2	27.844	27.824	28.737	29.320	29.308	30.179	17.937
PRIOR3	27.844	27.653	28.594	29.297	29.123	29.994	17.918

The first complete balanced panel supplies every latency sample. An earlier incomplete attempt failed an overly strict bit-exact replay check and is retained separately, not selected or rejected by speed. Fixed-input repeated GPU forwards located the prior’s first numerical differences in its first upsampling stage; the stem and ResNet blocks were identical in that diagnostic. The frozen nondeterministic cuDNN settings and all model weights remain unchanged. Corrected prior coordinates are checked in crop space using the existing network validation’s absolute component, 3×10^{-4} pixels, with zero relative allowance. Detector contracts and center8 remain bit-exact. Historical coordinates reproduce canonical pose metrics; current prior coordinates must preserve hypothesis and coverage. The final DEV accuracy tables still use the original frozen predictions.

After all balanced timing samples passed, the model-release guard prevented reuse of the same process for isolated memory measurements. The saved latency samples were retained unchanged; allocation was measured in a new process for each model. A further untimed coordinate replay audit belongs to that memory phase, not to the earlier timing samples. Per-sample timed numerical bounds passed, but individual drift magnitudes and OpenCV/inter-operation thread observations were not saved before the guard stopped the process. Those missing observations are explicitly distinguished from the separately recorded memory-phase settings. Recovery records and both original attempts are retained; no fastest session is chosen.

6 Controlled training and convergence limits

Parameter, memory and cache accounting

Model	Learned	Total	Add/R0(%)	Inf.MiB	Train MiB	s/update
R0	0	2615142	0.00	190.01	NM	NM
P1	18962	2634104	0.73	190.09	NM	0.0590
D1	19450	2634592	0.74	190.09	NM	0.0662
L1	19810	2634952	0.76	190.09	NM	NM
PRIOR1	68661641	71276783	2625.54	454.84	7779.02	0.4163

The shared R0 has 2,615,142 parameters. The recorded first cache pass processed 60,000 records in 1668.83 seconds; bound cache arrays occupy 68.677 GiB. Head fit time excludes that extraction. NM indicates an unavailable historical allocation measurement, not zero cost. R0 is not newly trained. CPU: 11th Gen Intel(R) Core(TM) i9-11900 @ 2.50GHz. Timed code fixes Torch intra-operation threads to 4. Inter-operation and OpenCV thread observations were not durably captured in the timed phase. In the separate fresh-process memory phase, observed settings were Torch intra-operation 4, inter-operation 8, OpenCV 1 (R0 process; all models are in the runtime JSON). These are not retrospectively labeled timed-phase observations. P and D share source order, feature evidence and historical optimizer/exposure settings, but their loss and output geometry differ. D’s small fixed-probe decrease is neither a convergence guarantee nor evidence of fundamental regression inferiority. The prior uses ImageNet TF-Slim ResNet152 initialization and the official loss/Adam semantics adapted to nine pallet points. Its sample order is exactly the existing P order and its nominal batch is sixteen. Source train/cal probes at step0 and every500 updates are diagnostic only. No checkpoint or wrapper is selected on DEV.

7 Operator and weight provenance

The official PoseFix source commit is 5556364bb0f43b0743a5fcd820de48f34b3d4360. Its MIT notice and the TensorFlow authors’ Apache-2.0 attribution are retained. The official TF1 CPU reference runs in an isolated Python3.7

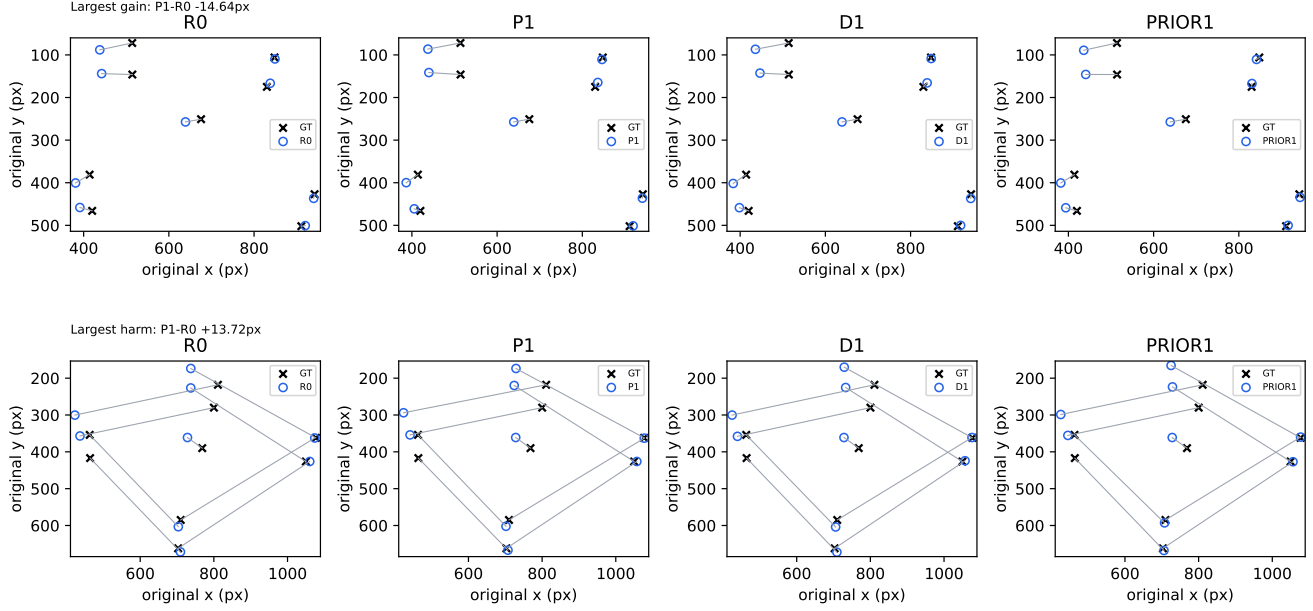


Figure 1: Actual improving and harming DEV cases: the lowest and highest per-frame median difference P1–R0, with lexical frame-ID ties. Same-frame GT and every plotted model use unchanged original coordinates and shared limits. Membership and raw sources are in `FIGURE_MANIFEST.json`. Images are not republished without verified image rights; these coordinate-only panels do not establish physical boundary correctness or independent generalization.

environment, and the PyTorch port uses the existing YOLO runtime without replacing that environment’s Torch, CUDA, or NumPy.

The reference initializes new weights using the official graph and transfers ImageNet RGB/backbone variables. Added pose channels remain at the official seeded initialization. The port maps convolution HWIO to OIHW, transpose convolution HWOI to IOHW, and every BN affine/moving variable. Missing and unexpected keys are explicitly checked. Full eval-coordinate and stage-feature comparisons use the actual source images and the same weights; scalar/loss-gradient and moving-statistics comparisons separately test training operators.

The graph’s nominal BN epsilon is not its effective value: the TF1 fused wrapper enforces 1.001×10^{-5} . The torch port matches this executed behavior. The Gaussian amplitude is 255 exactly once. Expectation coordinates are zero-based and scaled by 4 at the locked resolution. Bilinear target neighbors outside the grid are excluded and remaining mass normalized; nonfinite/outside-crop targets are masked in training, without altering evaluation membership. Raw RGB is explicitly the requested adaptation from the inspected BGR-preserving source loader.

The full batch 16 resource probe and actual parameter update are recorded separately from the three main fits. Main initialization and optimizer state do not reuse that smoke update. Checkpoints contain optimizer, Python/NumPy/CPU/CUDA RNG, step and exact order-prefix hash. The loader has no hidden random augmentation state; pending RGB reads can be reconstructed deterministically from the next recorded row.

8 Research history and interpretation

Prior Hough/line, active-adaptation, replay and robustness investigations motivated subsequent decisions, but do not become additional unseen validation sets. Old code, checkpoints, metrics and verdicts remain unchanged. This selection history means DEV is reused development evidence. The fixed model is not redesigned in response to an unfavorable prior comparison.

The baseline checkpoint is not purely synthetic-from-scratch: the recorded initialization checkpoint identifies COCO-pose training, despite a historical output directory containing the word scratch. P’s additional training uses synthetic labels only. ImageNet supervision for the added prior is separately disclosed. These distinctions prevent an overbroad claim that no real annotation contributed anywhere.

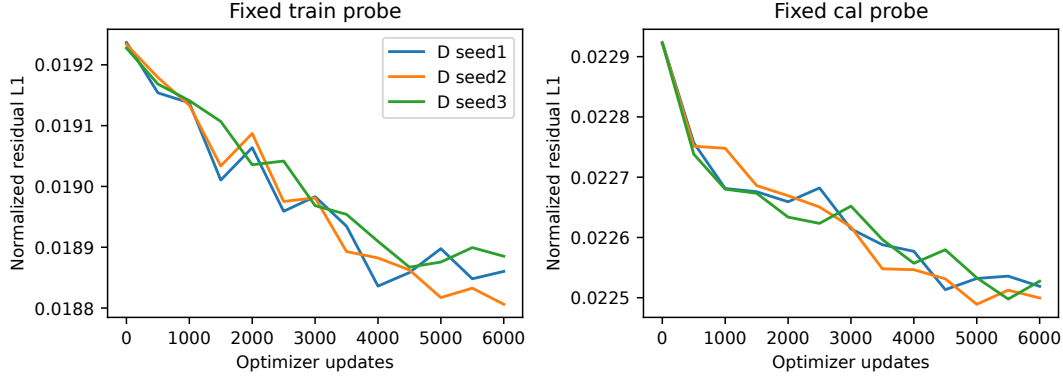


Figure 2: Already-completed D fixed probes. Their limited change is disclosed alongside the comparison, not suppressed. Loss magnitudes are not compared with P’s different objective.

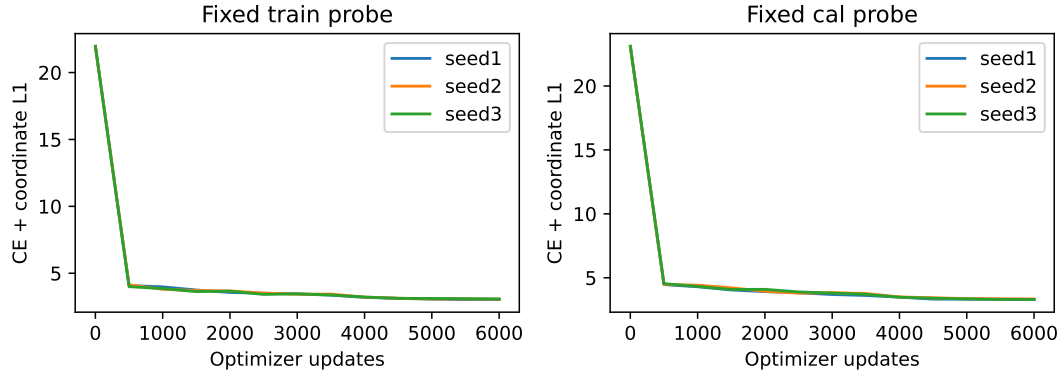


Figure 3: Prior fixed train/cal probes from the completed bounded fits. These curves did not select a checkpoint or extend training.

9 Independent data and annotation handoff

The actual collection and schema are in `HUMAN_INPUTS_REQUIRED.txt` and `CONFIRMATION_SCHEMA.json`. The starting plan is 12 separate sessions with 15 prespecified frames each; at least 30 frames receive two independent prediction-blinded annotations and blind adjudication. The count is not a power guarantee. Preserve original videos, timestamps, capture groups, camera/mount, calibration, actual object dimensions/identity and prior-use declarations. Changed image encoding or filename does not prove independence from an old scene.

Unreviewed provenance and an unfrozen comparator panel block confirmation only, not the completed development experiment. New-frame reference geometry must be bound to its camera/dimensions/index convention before a 6D claim. Human annotation, author identities, funding and consent cannot be fabricated by the CLI. These obligations are separate from compilation and code checks.

10 Three-estimator scope and outstanding evidence

The third application is a completed full-image, nine-landmark Simple-Baselines-derived ResNet18. Its frozen FULL estimator receives RGB and registered physical dimensions, and its primary transfer test adds the same image-feature local refiner without a second dimension input (P0). D0 is the matched direct-residual control. P5 and P5_CONSTANT form a separate dimension-input extension. All real-image outcomes are reused-development evidence. The completed accuracy and latency receipts are described below. Three measured frozen interfaces provide finite application evidence, not a universal model-agnostic theorem or an independent test.

Table 7: DOPE per-seed values; keypoint errors are pixels. Matched frames and pose fits have denominator 319; used/excluded GT sum to 2818. Excluded GT includes finite predictions on unmatched or incomplete frames, not only absent points. No failed population row is removed.

Arm	Med.	P90	Match	Used GT	Excl. GT	PCK10%	Pose n	T cm	R deg
DOPE	12.585	50.609	190	1710	1108	19.872	210	10.046	3.441
D1	7.857	47.334	190	1710	1108	35.983	210	8.352	2.920
D2	7.840	49.298	190	1710	1108	36.125	210	8.317	3.020
D3	7.903	48.831	190	1710	1108	36.054	210	8.300	3.115
P1	7.431	50.925	190	1710	1108	36.338	210	8.311	2.889
P2	7.422	50.433	190	1710	1108	37.048	210	8.496	2.893
P3	7.531	50.782	190	1710	1108	36.941	210	8.781	2.869

Table 8: DOPE paired session-bootstrap differences: first arm minus second. Negative values favor the first arm for these errors. An unavailable interval denotes incompatible or empty support, not a zero effect. These reused-DEV secondary intervals are unadjusted and exploratory.

Contrast	Keypoint px [95% CI]	T cm [95% CI]	R deg [95% CI]
P-DOPE	-5.124 [-6.304, -3.984]	-1.517 [-3.181, -0.407]	-0.558 [-1.082, -0.134]
D-DOPE	-4.718 [-6.035, -3.548]	-1.723 [-2.761, -0.856]	-0.423 [-1.082, -0.054]
P-D	-0.405 [-0.734, -0.061]	0.206 [-1.275, 1.062]	-0.135 [-0.566, 0.221]

11 DOPE extension: frozen interface and training lineage

For frozen DOPE on reused DEV, the conditional keypoint median is 12.585 pixels before refinement and 7.461 pixels for the mean of three P seed statistics; full-GT PCK10 is 19.872 and 36.775 percent, respectively. The tables retain missing predictions and unfavorable outcomes. The frozen checkpoint is `weights/backbone_dope_final_v1/run/final_net_epoch_0060.pth`; its SHA-256 begins `0de80490cb3b4f9b`, with the full digest in the protocol. The actual original training loop includes epochs 0 through 60. Original DOPE training uses random crop 400 and its recorded rotation, color, noise, and Torch FP32 normalization. The fixed inference adapter instead follows the existing `reflect100-resizeback-short400` path with its existing NumPy-to-FP32 normalization. A historical header value of 448 does not describe the executed crop. No equal-baseline-epoch or identical-training/inference-normalization claim is made.

The new adapter takes BGR input and returns explicit composed affine matrices, native/canvas/network sizes, semantic point validity, hull and confidence. Taps are post-ReLU 17/26; layer 16 alone is a convolution, not the feature used. Anisotropic inverse restoration adds the transformed displacement to the retained FP64 original point. This keeps a zero move exactly equal to the baseline and prevents a fresh FP32 reprojection from introducing approximately 10^{-5} -pixel roundoff. Fixed image-size, corner-index, GT-padding, camera and dimension contracts are required before scoring; there is no silent ID or geometry fallback.

The added P and D heads share 256/stride4 and 128/stride 8 evidence, 16-channel projections and 24 encoded channels. The backbone is in evaluation mode with gradients disabled. Newly initialized heads alone are optimized; an existing YOLO head is not a warm start. The source protocol fixes three seeds, 6,000 updates per head, batch 16 and six final fits. The AdamW learning rate is 10^{-3} , weight decay 10^{-4} , betas (0.9, 0.999), warmup 100, cosine terminal fraction 0.1, and gradient clip 5. These are head budgets, not convergence certificates for either family or equality of baseline-training cost.

Source prediction metadata is cached over the fixed source manifest; large full-resolution feature maps are not retained as a training cache. Online frozen VGG feature extraction therefore contributes actual training cost. Exact source image identities, usable rows, per-seed order hashes, final checkpoints, calibration temperatures and selected movement rules belong to the new receipts. The source mask can differ from the historical 55,915 usable YOLO rows. Nominal exposures must not be confused with successfully supervised corners; supported-frame loss reductions are retained.

Table 9: DOPE source heldout partition after wrapper freezing (1985 records). This source diagnostic uses its declared partial-point support and is not the complete-nine-point DEV statistic or independent TEST.

Arm	Med. px	P90 px	PCK10%	Used GT	Miss GT
DOPE	10.635	20.471	33.200	13167	4523
D1	4.558	16.948	57.614	13167	4523
D2	4.559	16.921	57.631	13167	4523
D3	4.560	16.894	57.513	13167	4523
P1	4.686	17.827	56.326	13167	4523
P2	4.693	17.886	56.490	13167	4523
P3	4.678	17.645	56.427	13167	4523

Table 10: Measured DOPE seed-1 desktop panel, 26 prespecified images with five repeats per arm after 20 warmups. Decoded RGB through real preprocessing/inference and canonical PnP; disk reads, loading and replay checks excluded. Each column is its own statistic: medians must not be added. All failure timings remain. This is a separate session from historical YOLO timing.

Arm	2D med. ms	PnP med. ms	Full med. ms	Full P90 ms	Pose OK/130
DOPE	60.786	1.625	62.205	73.310	85
DOPE+D1	63.888	1.671	65.038	76.349	85
DOPE+P1	63.361	1.678	64.924	76.220	85

12 DOPE results, missingness and paired uncertainty

The completed DOPE fits use 44063 usable training rows, 22034 learned parameters for P and 22522 for D. Six final fits each complete 6000 updates with 96000 nominal source exposures; no real images train these heads. P selects $\lambda = 1.0000$ and original-image-diagonal cap none. D selects $\lambda = 1.0000$ and original-image-diagonal cap none. P temperatures for seeds 1–3 are 1.00, 1.00, 1.00. The seven fixed arms are baseline DOPE and D1–D3/P1–P3. The complete population contains 319 positive images, 13 sessions and 2818 supervised keypoints. Every arm retains every population row, including failed or incomplete predictions. The conditional nine-point statistic requires the original box-IoU match and a finite nine-point prediction. Full-GT PCK counts excluded predictions as failures; separately labeled partial-point PCK is diagnostic only. No additional 2689-negative DOPE AP result is implied.

The evaluator reuses the canonical camera and dimension mapping, prediction-only choice of width or depth, and SQPnP followed by LM refinement. It does not let the reference pose choose a prediction hypothesis. Fewer than six finite corners, selector exceptions and failed PnP produce explicit failed rows. Pose medians use available fits and are always accompanied by coverage with denominator 319. A changed support prevents a conditional paired confidence interval rather than silently selecting a favorable intersection. Reference rotation/translation remain geometry-derived, with no new physical metrology.

Paired session and frame bootstrap reuse the existing seed 20260914 and 10000 draws. The family statistic averages per-seed pooled statistics, not predictions, and compares with one baseline. Source-selected wrappers do not use real outcomes, but the decision to run this second-estimator study followed earlier research on the same DEV population. The intervals are therefore exploratory and unadjusted. No independent TEST exists in this extension.

13 DOPE runtime and publication records

The completed panel contains 390 measured calls; coordinate replay passed at the declared absolute tolerance of 0.000000000100 pixels. Earlier failed attempts, if any, remain separately recorded and are not mixed into this completed summary. This runtime result does not establish accuracy or embedded-device throughput. The DOPE-only timing protocol records full DOPE+head RGB computation; no cached intermediate substitutes for the measured inference path. Seed 1 represents deployment without selecting the fastest seed. Keypoint and PnP-inclusive timings remain separate, and unavailable poses stay visible. A later integrated same-image panel supplies the main-text cross-estimator timing; the two runs are retained as different measurement sessions and their values are not interchanged.

The new experiment scripts are under `scripts/research/pallet_dope_refiner_20261001_v1`; the numer-

Table 11: Fixed epoch-10 synthetic calibration for the direct ResNet18 input arms. Soft errors decode spatial expectations; all arms use identical exposure and initialization.

Input arm	Loss	Med. px	P90 px	Match/1004
CONSTANT	0.43397	1.549	5.593	989
SHAPE	0.43304	1.570	5.516	992
FULL	0.43374	1.525	5.561	992

Table 12: Direct ResNet18 input-arm median 2D error (px). Wood is dimension-support extrapolation; GREEN150 and 0918 have fixed 1.10 x 1.10 x 0.15 m dimensions within each set and therefore do not identify a frame-varying dimension effect.

Reused development group	CONSTANT	SHAPE	FULL
DEV319	8.222	7.908	8.012
Plastic 110 x 130	7.587	7.483	7.732
Wood 80 x 59	9.430	8.492	8.234
GREEN150 square	5.785	5.601	4.826
0918 square (119)	6.584	6.497	6.298

ical records are under the matching `_docs/experiments/pallet_dope_refiner_20261001_v1` directory. The manuscript update script checks input bindings, population, arms and explicit completion before inserting results. `DOPE_RESULTS_BINDINGS.json` identifies inserted evidence, and `HISTORICAL_ASSETS.json` binds copied original manuscript assets. Pending evidence is stated explicitly rather than replaced by historical YOLO measurements. Build receipts certify typesetting only, not independent experimental verification or author approval.

14 ResNet18 direct estimator and dimension inputs

The ResNet18 FULL estimator and all twelve D0/P0/P5/P5_CONSTANT head fits completed. P0 is the primary third-backbone transfer arm; it improves central 2D development error and PCK10, while its translation interval includes zero. P5 is reported only as a separate dimension-conditioned-head extension because P5-P5_CONSTANT does not establish stable joint 2D/T/R improvement. The third estimator is explicitly a Simple-Baselines-derived ResNet18 full-image pallet9 adaptation, not PVNet, PoseFix or a renamed DOPE stage. The ImageNet file is `resnet18-f37072fd.pth`; the protocol records its full SHA-256. The 384×512 FP32 input uses aspect fitting and centered letterboxing without a predicted or ground-truth crop. Three 256-channel deconvolutions produce nine stride-four spatial distributions. Training minimizes 0.1 times normalized Gaussian spatial cross-entropy plus smooth-L1 error of the spatial expectation. The earlier 60-epoch masked-MSE pilot failed its fixed usability check and supplies none of the reported accuracy values.

The dimension vector is computed in fixed object axes and normalized from synthetic TRAIN only. FULL receives all five values through decoder FiLM, SHAPE receives the two ratios, and CONSTANT receives zeros through the same architecture. Each arm uses the same seed-42 initialization, source order, augmentation and fixed 10-epoch budget. The completed direct-estimator study trains CONSTANT, SHAPE and FULL arms for a fixed 10 epochs on 55,980 synthetic TRAIN images (34,990 updates and 559,800 exposures per arm). All share seed 42, ImageNet ResNet18 initialization, identical source order and photometric augmentation. FULL supplies normalized $\log W$, $\log D$, $\log H$, $\log(W/D)$ and $\log(H/\sqrt{WD})$ to a decoder FiLM block; SHAPE keeps only the two ratios and CONSTANT supplies zeros. Epoch 10 is fixed using no real-image selection.

The direct FULL-CONSTANT comparison is mixed: DEV median changes by -0.210 pixels, reconstructed-reference translation by $+1.269$ cm, and rotation by -0.760° . Plastic median worsens by 0.145 pixels. Wood improves by 1.196 pixels but its depth 0.59 m lies below the synthetic TRAIN minimum 0.818 m. GREEN150 and 0918 improve in aggregate, but each square set uses one constant dimension vector; the 0918 paired frame count is 59 improved versus 60 harmed with median change $+0.031$ pixels. These observations do not identify a stable causal dimension benefit.

The two pose summaries use different metric mappings and are intentionally not combined. The direct input report calls `scripts/research/pallet_dim_conditioned_p_v1/pose.py`, applies the fixed physical-frame conversion $R_{\text{physical}} = R_{\text{cf}}Q$, and then minimizes over its declared proper C2 group. The refiner evaluation calls the canonical MAIN `pose_evaluation_paths.score_pose_against_gt`, scores R_{cf} directly against `R_gt_representative`,

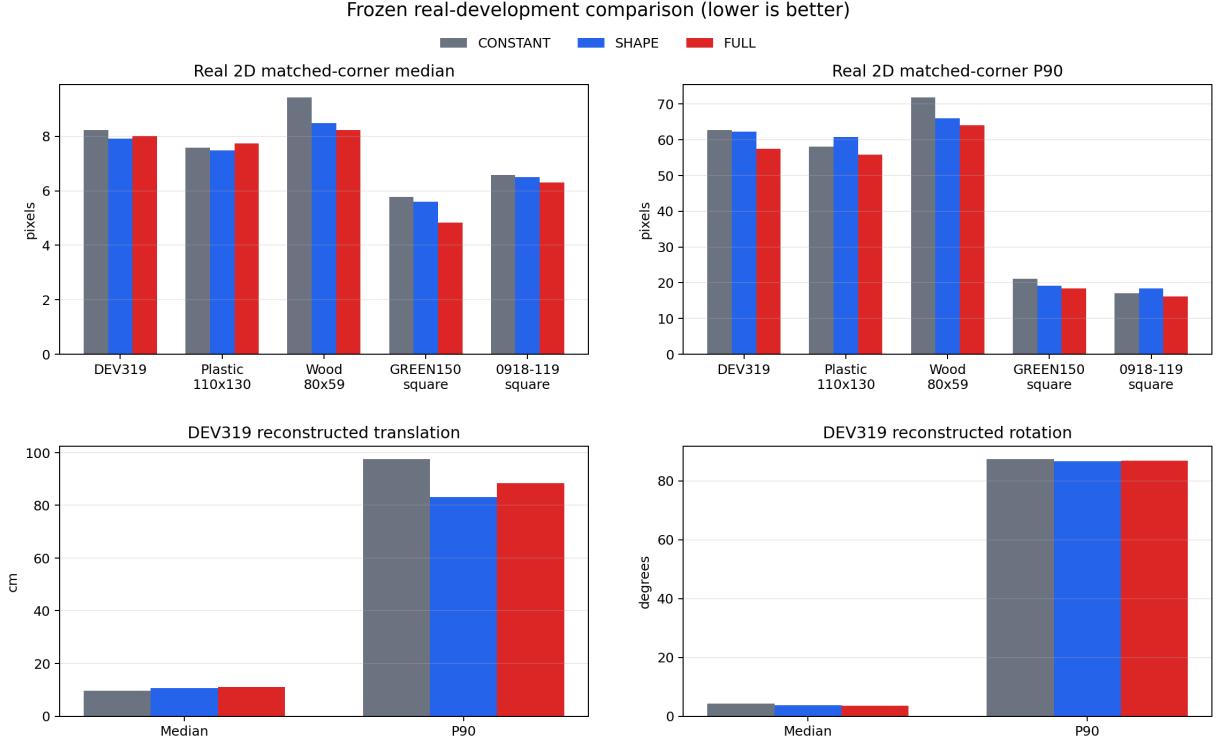


Figure 4: Separate direct-estimator input ablation. The lower panels use the direct experiment’s earlier proper-group pose scorer. They are separate from the canonical refiner before/after pose values reported below.

and reports width/depth parity separately as axis accuracy. Both use the same prediction-only width/depth selector and SQPnP/RefineLM, and both mark 234/319 frames axis-correct; the remaining 85 axis errors interact differently with the two rotation conventions. FULL translation consequently agrees to 0.0013 cm, while rotation, yaw, IoU and C2-aware corresponding-point ADD differ. For the same FULL checkpoint and numerically matched coordinate outputs (maximum point difference 0.001163 pixels), the direct report gives rotation 3.581° and C2-aware corresponding-point ADD AUC 0.31628; the canonical refiner report gives camera-facing-frame C2 rotation 3.252° and C2-aware corresponding-point ADD AUC 0.31296. This metric is not unrestricted nearest-neighbor ADD-S. Main-text before/after claims use only the latter internally consistent evaluator.

15 ResNet18 primary frozen-refiner transfer

FULL is frozen before head fitting. D0 and P0 use layer2/layer3 features at strides 8/16, the frozen predicted points and hull, and the same source-selection principles as the other estimator applications. Each head has three seeds, 6,000 updates and batch 16. P0, with 22,034 trainable parameters, is the primary third-backbone transfer arm. D0 is the 22,522-parameter matched direct residual control. Neither receives a new dimension vector; FULL’s earlier dimension conditioning is common to all rows.

The frozen FULL baseline yields 55,804 usable synthetic TRAIN rows. D0, P0, P5 and P5_CONSTANT each use three seeds of 6,000 AdamW updates at batch 16 (96,000 nominal exposures per fit) and zero real training images. Trainable counts are 22,522 for D0, 22,034 for P0, and 23,331 for each P5 arm. Synthetic selection fixes $\lambda = 1$ for every arm, a 0.01 image-diagonal cap only for D0, no cap for P0/P5, and temperature 1 for all probability heads.

P0 reduces conditional median and raises complete-denominator PCK10. Its session interval excludes zero for 2D median and canonical camera-facing-frame C2 rotation, while the translation interval includes zero. P90 changes from 52.054 to 52.086 pixels, so no tail-dominance claim is made. All arms retain 299 matched frames and 319/319 PnP coverage because the forced single-pallet FULL hull and output support are fixed.

Table 13: Primary ResNet18 arm by seed on reused DEV319. FULL is one frozen baseline; all rows preserve its 299 matched frames and 319/319 pose coverage. R is the canonical MAIN camera-facing-frame C2 rotation.

Arm	Med. px	P90 px	PCK10%	T cm	R-C2 deg
FULL	8.001	52.054	54.68	11.007	3.252
D0-S1	7.868	52.065	55.18	9.784	3.283
D0-S2	7.809	51.823	55.46	9.927	3.145
D0-S3	7.880	51.897	55.15	9.798	3.227
P0-S1	7.141	51.923	58.23	9.963	3.009
P0-S2	7.157	52.198	58.16	9.884	3.052
P0-S3	7.129	52.139	58.45	10.154	2.994

Table 14: Primary ResNet18 paired session-bootstrap differences on reused DEV: head minus FULL. Negative favors the head for 2D/T/R; positive favors it for PCK10. Intervals use 10,000 session resamples and are exploratory, unadjusted, and do not create an independent test.

Contrast	2D change px [95% CI]	PCK10 change pp [95% CI]	T change cm [95% CI]	R-C2 change deg [95% CI]
D0-FULL	-0.149 [-0.316,-0.048]	0.580 [0.242,0.983]	-1.171 [-1.388,0.149]	-0.034 [-0.216,0.062]
P0-FULL	-0.859 [-1.318,-0.621]	3.596 [2.515,4.986]	-1.007 [-1.668,0.187]	-0.234 [-0.502,-0.022]

16 Separate ResNet18 correction-head dimension extension

P5 adds a zero-initialized candidate-logit residual from the normalized five-value vector. P5_CONSTANT has the same module and initialization but supplies zeros. This is the incremental dimension-input contrast for the head. P5-P0 is only a package comparison because path and capacity also change; FULL is already dimension-conditioned in every arm.

17 ResNet18 integrated runtime and evidence limits

The integrated run measures FULL and one fixed seed for each head on the same image bytes used for YOLO and DOPE timing. FULL+P0 adds 2.436 ms to paired full-path median latency; P5 adds 3.407 ms. The desktop result includes model-specific preprocessing and all pose failures, but not file decoding, loading, Jetson export, concurrent throughput or energy.

All ResNet real-image panels are reused development data. The T/R reference is reconstructed from 2D annotation, camera intrinsics and registered dimensions rather than an independently instrumented 6D pose. Wood is dimension-support extrapolation and square-set dimensions do not vary. These limitations remain even though training, selection, evaluation and timing are complete.

18 Reproduction and access

From the repository root, use the recorded environment and `run.py status`, `run.py all`, or `run.py resume` under `scripts/research/pallet_sensors_submission_v1/`. Exact shell commands, downloads, locks and stage receipts are in `REPRODUCE.md`. Completion receipts are checked against actual input/output hashes, not accepted solely on a status string. Main uses no branch rewrite, force push or deletion of unrelated user changes.

Public materials contain code, small numerical results, plots and manuscript sources. Raw training/evaluation imagery, video, feature caches, model weights and the isolated environment remain excluded from public Git. Access restrictions and licenses must be reviewed before any separate sharing. No unverified speedup, annotation-cost reduction, or public-data right is inferred.

Table 15: Frozen synthetic heldout diagnostic after source-only rule selection (1,985 records). Head rows are means of three seed statistics; this reused source partition is not an independent real-image test.

Arm	Med. px	P90 px	PCK10%
FULL	2.805	9.499	89.723
D0 mean	2.694	9.504	89.721
P0 mean	2.585	9.510	89.710
P5 mean	2.585	9.500	89.674
P5_CONSTANT mean	2.585	9.506	89.674

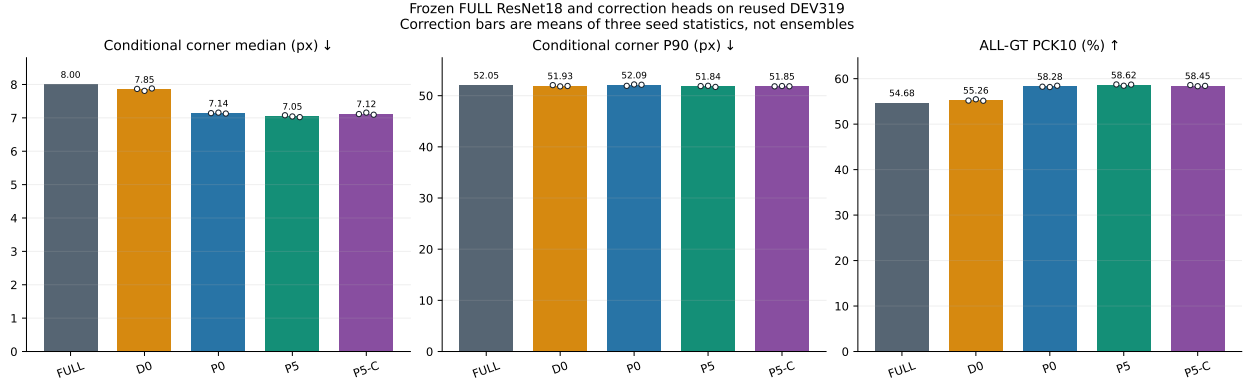


Figure 5: Frozen ResNet18 2D outcomes. Correction bars average three independently trained seed statistics; white circles are the seed values. P0 is the primary transfer arm.

Table 16: Separate ResNet18 correction-head dimension extension. P5_CONSTANT has the same P5 structure and initialization but replaces its five-value input with zeros. The direct FULL baseline is already dimension-conditioned in every row. Intervals are unadjusted reused-DEV session bootstraps.

Contrast or arm	2D med. px	PCK10%	T cm	R-C2 deg
P5_CONSTANT mean	7.122	58.45	9.982	3.009
P5 mean	7.048	58.62	9.911	2.982
P5-P5-C diff. [95% CI]	-0.074 [-0.116,0.019]	0.177 [0.011,0.397]	-0.071 [-0.115,0.071]	-0.027 [-0.067,0.025]

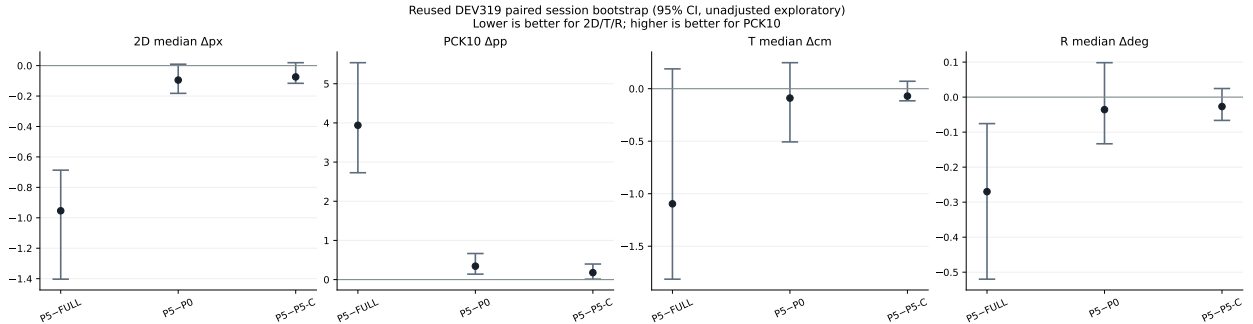


Figure 6: Exploratory reused-DEV paired intervals for the P5 extension. The P5-P5-C 2D, translation and rotation intervals include zero; a favorable PCK10 interval alone does not establish stable joint improvement.

Table 17: Integrated desktop timing for the ResNet18 arms. Each arm uses the same 26 decoded BGR images, 20 warmups, batch one, five repeats, and seed 1 for a learned head. PnP and full columns are separately summarized; their medians are not arithmetically additive.

Arm	2D med.	2D P90	Full med.	Full P90
FULL	7.835	10.432	9.453	12.146
D0-S1	10.104	13.410	11.886	14.838
P0-S1	10.520	13.565	12.193	14.982
P5-C-S1	11.442	13.790	13.132	15.228
P5-S1	11.302	13.977	12.963	15.342